
MARKET PULSE DETERMINISTIC SCORING ENGINE v2.6 (FINAL) 60 symbols across 7 clusters + 1 options post-processing layer.
All math is deterministic and clamp-verified. AI consumes JSON output for narrative ONLY. No math.

GENERAL RULES (apply everywhere)

1. All boundary conditions use $\geq$ and $\leq$ unless noted.
2. All "first match wins" lists evaluate from MOST EXTREME to LEAST EXTREME.
3. All cluster scores are clamped AFTER all adjustments.
4. All ratio calculations: if denominator = 0 or null, skip that component and mark STALE. Never compute $X/0$.
5. Integer overflow guard: if value > 9,007,199,254,740, set to null and mark STALE.
6. Pre-market stale detection: if $|\text{change}| < 0.001$ AND session is PRE_MARKET or OVERNIGHT AND indicator is in the known-stale list, force STALE.
7. Known-stale list: \$TNX, \$TYX, \$IRX, \$ADVN, \$DECN, \$TICK, \$TRIN, \$ADD, \$VVIX, \$VIX9D, \$VIX1D, \$VIX3M, \$SKEW, \$VXN, \$RVX, \$TICKI, \$STRINQ, \$ADDQ, \$ADVNQ, \$DECNQ, \$UVOL, \$DVOL, \$UVOLQ, \$DVOLQ, \$CPC, \$CPCE, \$CPCI, \$PCSPY, \$PCQQQ, \$PCIWM.

CLUSTER WEIGHTS (sum = 1.00) rates: 18% credit: 12% volLevel: 15% volTerm: 12% breadth: 18% riskAppetite: 15% macro: 10%

DATA QUALITY FRESH = value exists AND timestamp < 15 min STALE = value exists AND (timestamp $\geq$ 15 min OR pre-market stale triggered) MISSING = null / no data

CLUSTER CLAMP RANGES Clusters 1-6: [-2.0, +2.0] Cluster 7 (Macro): [-1.5, +1.5]

CLUSTER 1: RATES Weight: 18% | Clamp: [-2.0, +2.0] Inputs: \$TNX, \$TYX, \$IRX, /ZT, /ZF, /ZN, /ZB, /ZQ

PRE-MARKET GATE: If \$TNX is STALE AND \$TYX is STALE, skip Steps 2 and 3. Only Steps 1 (with fallback), 4, 5 run.

STEP 1 — Primary yield signal (\$TNX % change): (First match wins, most extreme first.) $\leq -1.0\% \rightarrow +1.0 \leq -0.5\% \rightarrow +0.75 \leq -0.1\% \rightarrow +0.5$

$-0.1\% \text{ AND } < +0.1\% \rightarrow 0 = +1.0\% \rightarrow -1.0 = +0.5\% \rightarrow -0.75 = +0.1\% \rightarrow -0.5$

PRE-MARKET FALLBACK (if \$TNX STALE, use /ZB): $/ZB \geq +0.3\% \rightarrow +0.75 /ZB \geq +0.1\% \rightarrow +0.5 > -0.1\% \text{ AND } < +0.1\% \rightarrow 0 /ZB \leq -0.3\% \rightarrow -0.75 /ZB \leq -0.1\% \rightarrow -0.5$

STEP 2 — Yield curve inversion (\$IRX vs \$TNX): (SKIP if either is STALE.) Both in Schwab x10 format. Compare raw values. $\$IRX > \$TNX \rightarrow -0.5$, flag: YIELD_CURVE_INVERTED $|\$IRX - \$TNX| \leq 1.0 \text{ (x10)} \rightarrow -0.25$, flag: YIELD_CURVE_FLATTENING $\$TNX > \$IRX \text{ by } > 5.0 \text{ (x10)} \rightarrow +0.25$ else $\rightarrow 0$

STEP 3 — Curve shape (\$TYX vs \$TNX % changes): (SKIP if either is STALE.) $\text{curveDiff} = \text{tnxChange} - \text{tyxChange}$

Scenario	curveDiff	Score	
falling, $> +0.1$	$+0.25$ (bull TYX falls MORE steepener)		Both
falling, < -0.1	-0.25 (bear TNX falls MORE flattener)		Both
rising, < -0.1	-0.25 (bear TYX rises MORE steepener)		Both
rising, $> +0.1$	-0.25 (bear TNX rises MORE flattener)		Both
Opposite directions	any $-0.25 + \text{flag}$ (both $ \text{chg} \geq 0.1\%$) CURVE_DISTORTION		

other diff <=0.1 0				All
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Only bull steepener (both falling, long end leads) is positive. All other meaningful curve moves are bearish.

STEP 4 — Bond futures confirmation (/ZT, /ZF, /ZN, /ZB): Count futures with positive price change (rallying). ratesIntermediate = running sum of Steps 1+2+3.

ratesIntermediate > 0: >= 3 rallying → +0.25 1-2 rallying → 0 0 rallying → -0.25 ratesIntermediate < 0: <= 1 rallying → -0.25 2-3 rallying → 0 4 rallying → +0.25 ratesIntermediate = 0: → 0

STEP 5 — Fed Funds (/ZQ % change):

| = +0.02% → +0.25 (dovish) <= -0.02% → -0.25 (hawkish) else → 0

ratesRaw = Step1 + Step2 + Step3 + Step4 + Step5 ratesScore = clamp(ratesRaw, -2.0, +2.0)

Headlines:

| = +1.5 → "Yields falling sharply — strong risk-on" = +0.5 → "Yields declining — supportive for equities" <= -1.5 → "Yields surging — recession risk elevated" <= -0.5 → "Yields rising — headwind for equities" else → "Yields stable — neutral signal"

CLUSTER 2: CREDIT Weight: 12% | Clamp: [-2.0, +2.0] Inputs: HYG, LQD, IEF, TLT

HYG (% change, first match wins):

| = +0.5% → +2.0 = +0.2% → +1.0 -0.2% AND < +0.2% → 0 <= -0.5% → -2.0 <= -0.2% → -1.0

LQD (% change):

| = +0.3% → +0.5 <= -0.3% → -0.5 else → 0

IEF (% change) — DIRECTION-DEPENDENT: IF HYG >= +0.1%: IEF >= +0.3% → +0.25 IEF <= -0.3% → -0.25 else → 0 IF HYG <= -0.1%: IEF >= +0.3% → 0 (flight-to-safety, not credit positive) IEF <= -0.3% → -0.25 (everything selling) else → 0 IF HYG flat → 0

TLT (% change) — DIRECTION-DEPENDENT: IF HYG >= +0.1%: TLT >= +0.5% → +0.5 TLT <= -0.5% → -0.5 else → 0 IF HYG <= -0.1%: TLT >= +0.5% → 0 (flight-to-safety) TLT <= -0.5% → -0.5 (everything selling) else → 0 IF HYG flat → 0

creditRaw = HYG + LQD + IEF + TLT

Flags (informational, no score change): HYG >= +0.2% AND (IEF >= +0.2% OR TLT >= +0.2%) → CREDIT_FLIGHT_TO_SAFETY TLT >= +0.5% AND HYG <= -0.3% → CREDIT_STRESS_CONFIRMED

creditScore = clamp(creditRaw, -2.0, +2.0)

Headlines:

| = +1.5 → "Credit surging — strong risk-on" = +0.5 → "Credit improving — supportive backdrop" <= -1.5 → "Credit deteriorating — risk-off alert" <= -0.5 → "Credit weakening — caution warranted" else → "Credit stable — neutral signal"

CLUSTER 3: VOL LEVEL Weight: 15% | Clamp: [-2.0, +2.0] Inputs: \$VIX, \$VVIX, \$VXN, \$RVX, \$OVX, \$GVZ, /VIX Experimental: \$TYVIX, \$MOVE

STEP 1 — \$VIX absolute level: < 15 → +0.5 < 20 → +0.25 < 25 → 0 < 30 → -0.5 < 35 → -1.0

| = 35 → -1.5

STEP 2 — \$VIX intraday % change (first match): <= -5% → +1.5 <= -3% → +0.75 <= -1% → +0.25

| -1% AND < +1% → 0 = +5% → -1.5 = +3% → -0.75 = +1% → -0.25

STEP 3 — \$VVIX % change: <= -3% → +0.5

| = +3% → -0.5 else → 0

STEP 4 — Cross-vol (\$VXN, \$RVX vs \$VIX): VXN >= VIX+5 AND RVX >= VIX+5 → -0.5, flags: TECH_VOL_ELEVATED, SMALLCAP_VOL_ELEVATED VXN >= VIX+5 only → -0.25, flag: TECH_VOL_ELEVATED RVX >= VIX+5 only → -0.25, flag: SMALLCAP_VOL_ELEVATED VXN < VIX AND RVX < VIX → +0.25 else → 0

STEP 5 — Commodity vol (\$OVX, \$GVZ): OVX >= 60 → -0.5 OVX >= 45 → -0.25 OVX >= 30 → 0 (explicit neutral) GVZ >= 25 → -0.25 (additive) OVX < 30 AND GVZ < 15 → +0.25 else → 0

STEP 6 — /VIX futures vs \$VIX spot: spread = /VIX - \$VIX (Contango = bullish, backwardation = bearish. Consistent with Vol Term Component E.) spread >= +2.0 → +0.5 (steep contango) spread >= +0.5 → +0.25 (mild contango) |spread| < 0.5 → 0 spread <= -2.0 → -0.5 (steep backwardation) spread <= -0.5 → -0.25 (mild backwardation)

STEP 7 — EXPERIMENTAL (skip if MISSING): \$MOVE >= 130 → -0.25 \$MOVE < 80 → +0.25 \$TYVIX >= 10 → -0.25

volLevelRaw = Steps 1-7 summed volLevelScore = clamp(volLevelRaw, -2.0, +2.0)

Direction:

= +0.25 → COMPRESSING <= -0.25 → EXPANDING else → STABLE

CLUSTER 4: VOL TERM STRUCTURE Weight: 12% | Clamp: [-2.0, +2.0] Final = AVERAGE of active components. Min 2 required.
Inputs: \$VIX1D, \$VIX9D, \$VIX3M, \$SKEW, /VIX

COMPONENT A — VIX1D / VIX9D ratio (skip if either STALE): < 0.85 → +2.0 < 0.95 → +1.0 < 1.05 → 0 < 1.15 → -1.0

= 1.15 → -2.0

COMPONENT B — VIX9D / VIX ratio: < 0.90 → +2.0 < 0.95 → +1.0 < 1.05 → 0 < 1.10 → -1.0

= 1.10 → -2.0

COMPONENT C — VIX / VIX3M ratio: < 0.85 → +2.0 < 0.95 → +1.0 < 1.05 → 0 < 1.15 → -1.0

= 1.15 → -2.0

COMPONENT D — \$SKEW absolute:

= 150 → -2.0 = 140 → -1.0 = 130 → 0 = 115 → +1.0 < 115 → +2.0

COMPONENT E — /VIX vs \$VIX (first match, most extreme): spread = /VIX - \$VIX spread >= +2.0 → +2.0 (steep contango) spread >= +0.5 → +1.0 (mild contango) |spread| < 0.5 → 0 spread <= -2.0 → -2.0 (steep backwardation) spread <= -0.5 → -1.0 (mild backwardation)

activeCount = non-STALE, non-MISSING components IF activeCount < 2: volTermScore = 0, dataQuality = STALE ELSE:

volTermRaw = sum / activeCount volTermScore = clamp(volTermRaw, -2.0, +2.0)

Direction:

= +0.25 → CONTANGO <= -0.25 → INVERTED else → FLAT

Flag: ALL of A,B,C active AND negative, AND D active AND negative → FULL_TERM_INVERSION

CLUSTER 5: BREADTH Weight: 18% | Clamp: [-2.0, +2.0] Inputs: \$TICK, \$TRIN, \$ADD, \$ADV, \$DECN, \$UVOL, \$DVOL, \$TICKI, \$TRINQ, \$ADDQ, \$ADVQ, \$DECNQ, \$UVOLQ, \$DVOLQ

GUARDS: Denominator = 0 → skip component. Value > 9e12 → null + STALE. Both exchanges MISSING → score 0, MISSING.

NYSE (average of available components): \$TICK: >= +1000 → +2.0, >= +600 → +1.0, >= +200 → +0.5, > -200 → 0, >= -600 → -0.5, >= -1000 → -1.0, < -1000 → -2.0 \$TRIN: < 0.80 → +2.0, < 1.00 → +1.0, < 1.20 → -1.0, >= 1.20 → -2.0 \$ADD: >= +1500 → +2.0, >= +800 → +1.0, >= +200 → +0.5, > -200 → 0, >= -800 → -0.5, >= -1500 → -1.0, < -1500 → -2.0 \$ADV/\$DECN (skip if \$DECN=0): >= 3.0 → +2.0, >= 2.0 → +1.0, >= 0.5 → 0, >= 0.33 → -1.0, < 0.33 → -2.0 \$UVOL/\$DVOL (skip if \$DVOL=0 or overflow): >= 4.0 → +2.0, >= 2.0 → +1.0, >= 0.5 → 0, >= 0.25 → -1.0, < 0.25 → -2.0 nyseScore = sum of available / count of available

NASDAQ: Same thresholds for \$TICKI, \$TRINQ, \$ADDQ, \$ADVQ/\$DECNQ, \$UVOLQ/\$DVOLQ.

breadthRaw = IF(both) AVERAGE(nyse, nasdaq) ELIF(NYSE only) nyse ELIF(NASDAQ only) nasdaq ELSE 0

Exchange divergence (both available): |nyseScore - nasdaqScore| >= 1.5 → breadthRaw -= 0.25, flag: EXCHANGE_DIVERGENCE

breadthScore = clamp(breadthRaw, -2.0, +2.0)

Direction: >= +0.5 → BROAD_PARTICIPATION, <= -0.5 → BROAD_DETERIORATION, else → MIXED

CLUSTER 6: RISK APPETITE Weight: 15% | Clamp: [-2.0, +2.0] Inputs: /ES, /NQ, /YM, /RTY, SPY, QQQ, IWM

STEP 1 — Futures directional: futuresAvg = avg of /ES, /NQ, /YM, /RTY changes allPositive = all 4 >= 0, allNegative = all 4 <= 0
allPos AND avg >= +1.0% → +1.5 allPos AND avg >= +0.5% → +1.0 allPos → +0.5 allNeg AND avg <= -1.0% → -1.5 allNeg AND avg
<= -0.5% → -1.0 allNeg → -0.5 mixed → 0

STEP 2 — /RTY leadership: rtyDiff = /RTY - /ES change

| = +0.5% → +0.5, <= -0.5% → -0.5, else → 0

STEP 3 — /NQ concentration: nqDiff = /NQ - avg(/ES, /YM, /RTY)

| = +1.0% → -0.25, <= -1.0% → -0.25, else → 0

STEP 4 — Cash confirmation: |SPY - /ES| >= 0.3% → -0.25 IWM - SPY >= +0.5% → +0.25 QQQ outperforming SPY >= 1.0% AND
IWM underperforming → -0.25 none trigger → 0

riskAppRaw = Steps 1-4 riskAppScore = clamp(riskAppRaw, -2.0, +2.0)

Headlines:

| = +1.5 → "Broad equity rally — risk appetite strong" = +0.5 → "Futures positive — risk-on posture" <= -1.5 → "Broad equity
selloff — risk-off" <= -0.5 → "Futures under pressure — defensive tone" else → "Futures mixed — no directional conviction"

CLUSTER 7: MACRO Weight: 10% | Clamp: [-1.5, +1.5] Inputs: /GC, /CL, /BZ, /HG, /DX, /6E, /6J

/GC Gold (first match): <= -1.0% → +0.5, <= -0.5% → +0.25,

| -0.5% AND < +0.5% → 0, = +1.0% → -0.5, >= +0.5% → -0.25

/HG Copper (first match):

| = +1.5% → +0.5, >= +0.5% → +0.25, -0.5% AND < +0.5% → 0, <= -1.5% → -0.5, <= -0.5% → -0.25

Oil avg = (/CL + /BZ) / 2 (first match):

| = +3.0% → -0.5, >= +1.5% → -0.25, = 0% → +0.25, >= -1.5% → +0.25, = -3.0% → -0.25, < -3.0% → -0.5

/DX Dollar (first match): <= -0.5% → +0.5, <= -0.2% → +0.25,

| -0.2% AND < +0.2% → 0, = +0.5% → -0.5, >= +0.2% → -0.25

/6J Yen (rising = risk-off, first match):

| = +0.5% → -0.5, >= +0.2% → -0.25, -0.2% AND < +0.2% → 0, <= -0.5% → +0.5, <= -0.2% → +0.25

/6E Euro (confirmation): Both /6E and /DX same direction (each >= 0.1%) → -0.25, flag: FX_DISLOCATION else → 0

macroRaw = sum all macroScore = clamp(macroRaw, -1.5, +1.5)

Flags: /GC >= +0.5% AND /DX >= +0.3% → GOLD_DOLLAR_SPIKE /HG <= -0.5% AND oilAvg <= -0.5% AND /6J >= +0.2% →
GLOBAL_GROWTH_SCARE

Headlines:

| = +1.0 → "Global risk-on — dollar weak, copper strong" = +0.5 → "Macro supportive — mild risk-on flows" <= -1.0 → "Global
risk-off — stress across FX/commodities" <= -0.5 → "Macro headwinds — defensive flows building" else → "Macro backdrop
neutral"

COMPOSITE SCORE

$\text{composite} = (\text{rates} \times 0.18) + (\text{credit} \times 0.12) + (\text{volLevel} \times 0.15) + (\text{volTerm} \times 0.12) + (\text{breadth} \times 0.18) + (\text{riskApp} \times 0.15) + (\text{macro} \times 0.10)$

Theoretical max: ± 1.95

COMPOSITE MOMENTUM DELTA

Store last 20 composite scores with timestamps (4-hour TTL).

$\text{previousComposite} = \text{most recent prior generation}$ $\text{timeDeltaMinutes} = \text{minutes between current and previous}$

IF no previous exists: $\text{delta} = \text{null}$, $\text{label} = \text{"FIRST_READ"}$ ELIF $\text{timeDeltaMinutes} > 120$: $\text{delta} = \text{null}$, $\text{label} = \text{"STALE_COMPARISON"}$
ELIF session changed (e.g., yesterday close $\rightarrow$ today pre-market): $\text{delta} = \text{null}$, $\text{label} = \text{"SESSION_GAP"}$ ELSE: $\text{delta} = \text{currentComposite} - \text{previousComposite}$ $\text{delta} \geq +0.30 \rightarrow \text{"ACCELERATING"}$ $\text{delta} \geq +0.10 \rightarrow \text{"IMPROVING"}$ $\text{delta} > -0.10 \rightarrow \text{"STABLE"}$ $\text{delta} > -0.30 \rightarrow \text{"FADING"}$ $\text{delta} \leq -0.30 \rightarrow \text{"DETERIORATING"}$

Momentum flags: Bullish bias + DETERIORATING $\rightarrow$ BULLISH_MOMENTUM_FADING Bullish bias + ACCELERATING $\rightarrow$ BULLISH_MOMENTUM_BUILDING Bearish bias + DETERIORATING $\rightarrow$ BEARISH_MOMENTUM_BUILDING Bearish bias + IMPROVING $\rightarrow$ BEARISH_MOMENTUM_FADING

CONFIDENCE (0-100)

$\text{base} = \text{MIN}(95, |\text{composite}| \times 85)$

ZERO-BOUND GUARD: IF $|\text{composite}| < 0.01$: $\text{agreementMult} = 1.0$ (skip check) ELSE: Agreement multiplier (start 1.0): Each cluster OPPOSITE composite direction: $|\text{score}| \geq 1.0 \rightarrow -0.15$ (strongly opposing) $|\text{score}| \geq 0.5 \rightarrow -0.10$ (moderately opposing) Floor: 0.40

Data quality multiplier (start 1.0): Each MISSING: -0.15 Each STALE: -0.05 Floor: 0.50

$\text{rawConfidence} = \text{ROUND}(\text{base} \times \text{agreementMult} \times \text{dataQualityMult})$

STRUCTURAL CAPS (applied in order): $\text{hasDivergence} = \text{true} \rightarrow \text{MIN}(\text{rawConfidence}, 60)$ volTerm INVERTED AND volLevel EXPANDING $\rightarrow \text{MIN}(\text{conf}, 50)$ $\text{freshClusterCount} < 5 \rightarrow \text{MIN}(\text{conf}, 40 + \text{freshCount} \times 10)$

GAP-RISK GUARD: IF $\text{session} = \text{PRE_MARKET}$ AND $\text{riskAppScore} \geq +1.0$ AND breadth dataQuality is STALE or MISSING: $\text{confidence} -= 10$ flag: GAP_RISK_UNCONFIRMED

$\text{confidence} = \text{clamp}(\text{finalResult}, 0, 100)$

Denominator: $\text{maxPossible} = \text{ROUND}(100 \times \text{dataQualityMult})$ Display: "confidence / maxPossible"

BIAS LABEL (first match wins)

$\text{confidence} < 20 \rightarrow \text{NO_EDGE}$ $\text{composite} \geq +1.50$ AND $\text{confidence} \geq 75 \rightarrow \text{STRONGLY_BULLISH}$ $\text{composite} \geq +0.75 \rightarrow \text{MODERATELY_BULLISH}$ $\text{composite} \geq +0.30 \rightarrow \text{SLIGHTLY_BULLISH}$ $\text{composite} > -0.30 \rightarrow \text{NEUTRAL}$ $\text{composite} \geq -0.75 \rightarrow \text{SLIGHTLY_BEARISH}$ $\text{composite} \geq -1.50 \rightarrow \text{MODERATELY_BEARISH}$ $\text{composite} < -1.50$ AND $\text{confidence} \geq 75 \rightarrow \text{STRONGLY_BEARISH}$ else $\rightarrow \text{MODERATELY_BEARISH}$

TODAY'S EDGE

$\text{confidence} < 20 \rightarrow \text{NO_EDGE}$ $\text{composite} \geq +0.5 \rightarrow \text{BULLISH_EDGE}$ $\text{composite} \leq -0.5 \rightarrow \text{BEARISH_EDGE}$ else $\rightarrow \text{NEUTRAL_EDGE}$

DIVERGENCE DETECTION

Primary (threshold ± 0.75): Any cluster $\geq +0.75$ AND any cluster ≤ -0.75 : hasDivergence = true riskState forced to REDUCED confidence capped at 60

Secondary flags: YIELD_CURVE_INVERTED, YIELD_CURVE_FLATTENING, CURVE_DISTORTION, CREDIT_FLIGHT_TO_SAFETY, CREDIT_STRESS_CONFIRMED, TECH_VOL_ELEVATED, SMALLCAP_VOL_ELEVATED, FULL_TERM_INVERSION, EXCHANGE_DIVERGENCE, FX_DISLOCATION, GOLD_DOLLAR_SPIKE, GLOBAL_GROWTH_SCARE, GAP_RISK_UNCONFIRMED, BULLISH_MOMENTUM_FADING, BULLISH_MOMENTUM_BUILDING, BEARISH_MOMENTUM_BUILDING, BEARISH_MOMENTUM_FADING, SESSION_GAP

STRUCTURAL REGIME

freshClusterCount < 3 $\rightarrow$ NO_READ composite $\geq +0.5 \rightarrow$ RISK_ON composite $\leq -0.5 \rightarrow$ RISK_OFF else $\rightarrow$ TRANSITION

RISK STATE (first match wins)

confidence < 20 $\rightarrow$ NO_TRADE hasDivergence = true $\rightarrow$ REDUCED |composite| ≥ 1.0 AND confidence $\geq 70 \rightarrow$ PRESS |composite| $\geq 0.5 \rightarrow$ NORMAL else $\rightarrow$ REDUCED

SIZE RECOMMENDATION (derived from Risk State)

PRESS $\rightarrow$ FULL_SIZE NORMAL $\rightarrow$ FULL_SIZE REDUCED $\rightarrow$ HALF_SIZE NO_TRADE $\rightarrow$ NO_TRADE

RISK STATE REASONING (NEW in v2.6)

Populate an array of strings explaining WHY the risk state was determined. Passed to AI and available for UI tooltips.

Examples: ["hasDivergence: rates bullish vs vol bearish", "confidence capped at 60"]

["composite +1.2 with confidence 82", "no divergence, all clusters aligned"]

["confidence 18 below NO_TRADE threshold", "3 clusters STALE, data insufficient"]

["pre-market gap risk: futures +1.4% unconfirmed", "breadth data stale"]

OPTIONS POST-PROCESSING LAYER Runs AFTER composite. Not a weighted cluster. Inputs: \$CPC, \$CPCE, \$CPCI, \$PCSPY, \$PCQQQ, \$PCIWM, IVR If put/call unavailable, skip sentiment. Still output IVR and vol-safety.

STEP 1 — IVR (0-100): Source: 52-week IV percentile from options chain. Fallback VIX proxy: VIX $\geq 35 \rightarrow 90$, $\geq 30 \rightarrow 75$, $\geq 25 \rightarrow 55$, $\geq 20 \rightarrow 40$, $\geq 18 \rightarrow 30$, $\geq 15 \rightarrow 20$, $< 15 \rightarrow 10$

IVR $\geq 50 \rightarrow$ SELL_PREMIUM IVR 25-49 $\rightarrow$ NEUTRAL_IV IVR < 25 $\rightarrow$ BUY_PREMIUM

STEP 2 — Broad sentiment (\$CPCE):

| = 0.90 $\rightarrow$ EXTREME_FEAR = 0.75 $\rightarrow$ ELEVATED_FEAR = 0.55 $\rightarrow$ NEUTRAL < 0.55 $\rightarrow$ EXTREME_GREED

STEP 3 — Institutional (\$CPCI):

| = 1.5 $\rightarrow$ INSTITUTIONS_HEDGING_HARD = 1.1 $\rightarrow$ INSTITUTIONS_HEDGING = 0.8 $\rightarrow$ INSTITUTIONAL_NEUTRAL < 0.8 $\rightarrow$ INSTITUTIONS_COMPLACENT

STEP 4 — Ticker flags: \$PCSPY $\geq 1.0 \rightarrow$ SPY_FEAR_ELEVATED \$PCQQQ $\geq 1.0 \rightarrow$ QQQ_FEAR_ELEVATED \$PCIWM $\geq 1.2 \rightarrow$ IWM_FEAR_ELEVATED \$PCSPY < 0.6 $\rightarrow$ SPY_COMPLACENT \$PCQQQ < 0.5 $\rightarrow$ QQQ_COMPLACENT

STEP 5 — AVOID_SHORT_PREMIUM (triggers if ANY true): a) volTerm INVERTED AND volLevel EXPANDING b) \$VVIX change $\geq +5\%$ c) CREDIT_STRESS_CONFIRMED flag active

STEP 6 — Trade type (first match): AVOID_SHORT_PREMIUM active: $\rightarrow$ "Vol/credit unsafe. Buy spreads only or sit out."

EXTREME_FEAR AND composite ≥ 0.0 : → "Capitulation setup — mean-reversion long"

EXTREME_GREED AND composite $\leq +0.5$: → "Complacency warning — reduce or hedge longs"

INSTITUTIONS_HEDGING_HARD AND bias BULLISH: → "Institutional hedging elevated — defined-risk only"

STRONGLY_BULLISH + SELL_PREMIUM: → "Sell puts or bull put spreads. Defined-risk recommended." STRONGLY_BULLISH + BUY_PREMIUM: → "Long calls or bull call spreads." MODERATELY_BULLISH + SELL_PREMIUM: → "Bull put spreads on preferred tickers." MODERATELY_BULLISH + BUY_PREMIUM: → "Bull call spreads on preferred tickers." SLIGHTLY_BULLISH: → "Lean bullish — reduced size, defined-risk only."

STRONGLY_BEARISH + SELL_PREMIUM: → "Sell calls or bear call spreads. Defined-risk recommended." STRONGLY_BEARISH + BUY_PREMIUM: → "Long puts or bear put spreads." MODERATELY_BEARISH + SELL_PREMIUM: → "Bear call spreads on preferred tickers." MODERATELY_BEARISH + BUY_PREMIUM: → "Bear put spreads on preferred tickers." SLIGHTLY_BEARISH: → "Lean bearish — defined-risk only, small size."

NEUTRAL/NO_EDGE + SELL_PREMIUM + NOT_AVOID: → "No directional edge. Iron condors if IV elevated." NEUTRAL/NO_EDGE otherwise: → "No conviction. Sit out."

LEVELS TO WATCH

\$VIX : current ± 2 pts \$TNX : current ± 0.10 (x10) \$IRX : vs \$TNX (inversion trigger) HYG : current - 0.3% /ES : current $\times 0.99 / 1.01$
\$ADD : 0 line \$TRIN : 1.0 line /HG : current - 1.0% /DX : current + 0.5%

AI JSON PAYLOAD

```
{ engineVersion: "v2.6", composite, biasLabel, confidence, confidenceDenominator, structuralRegime, riskState,
riskStateReasoning: string[], todaysEdge, sizeRecommendation, hasDivergence, divergenceDetail: { bullishClusters[],
bearishClusters[] }, secondaryFlags[], clusterScores: { rates, credit, volLevel, volTerm, breadth, riskApp, macro }, clusterRaws: {
rates, credit, volLevel, volTerm, breadth, riskApp, macro }, clusterHeadlines: { rates, credit, volLevel, volTerm, breadth, riskApp,
macro }, clusterDirections: { volLevel, volTerm, breadth }, clusterDataQuality: { rates, credit, volLevel, volTerm, breadth, riskApp,
macro }, breadthDetail: { nyseScore, nasdaqScore, exchangeDivergence: bool }, momentum: { delta: number | null, label: string,
previousComposite: number | null, previousTimestamp: string | null, timeDeltaMinutes: number | null }, optionsLayer: { ivRank,
premiumDirection, broadSentiment, institutionalHedging, tickerFlags[], avoidShortPremium: bool, avoidReason: string | null,
tradeTypeRecommendation }, levelsToWatch: { vix, tn timer, irx, hyg, es, add, trin, hg, dx }, dataQuality: { freshCount, staleCount,
missingCount, missingClusters[], staleClusters[], experimentalAvailable: { tyvix, move } } }
```

RULE: AI writes narrative and explains the score. AI does NOT compute, modify, or override any number. If AI output contains any number not in the JSON, that number is WRONG and must be rejected by frontend.
